1. Abstract

In many nonlinear dynamic control problems, the optimal value function is the solution of the Hamilton-Jacobi-Bellman PDEs, which are often high-dimensional. The data driven approach with gradient augmentation has been used to solve the high-dimensional PDEs in order to circumvent the curse of dimensionality. In [1] (Kunisch et al., 2021), a data set consisting of state-adjoint-control triples is generated by solving the open-loop problem provided by the Pontryagin Maximum Principle. The optimal value function is in turn approximated by hyperbolic cross basis functions with l^1 penalty to enhance the sparsity. The gradient augmentation combined with hyperbolic cross basis is proven to be very effective to overcome the curse of dimensionality.

In terms of different approximation methods besides the hyperbolic basis functions, there is a growing literature on solving the nonlinear optimal control problem using neural networks. Although the number of neurons may not serve as a control parameter of the model capacity [10], sparsity is crucial for the computational efficiency in many problems. A sparse learning procedure usually adopts convex reformulation of the problem and employ a sparse promoting convex penalty term on the outer weights, a popular choice is the l^1 function (LASSO). However it was indicated in [12] that the convex penalty terms are not able to effectively eliminate the redundancy, especially when the neurons are clustering in a small neighborhood. A nonconvex sparse learning method is proposed by the author.

In the paper [12](Pieper et al., 2022), the authors choose ReLU as the activation function. The ReLU functions enjoy many nice properties such as Lipschitz continuity and positive homogeneity, which are essential to prove the optimality and for the penalty term to promote the sparsity. The problem is that the ReLU activation functions are not smooth, which may suffer from exploding gradient. In this thesis we propose a different activation function that may circumvent this problem. And as in [1] we incorporate the gradient information in the cost function to obtain optimal convergence.

2. Introduction

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$
(P) \quad \begin{aligned}
&\inf_{u \in L^2(t_0,T,\mathbb{R}^k)} J(u;t_0,y) := \int_{t_0}^{T} l(y(t)) + \beta |u(t)|^2 dt, \quad \beta > 0 \\
&\text{s.t.} \\
&\dot{y} = f(y) + g(y)u, \\
&y(t_0) = x,
\end{aligned}
$$

where we assume $y(t) \in \mathbb{R}^d, u(t) \in \mathbb{R}^k$, $l \in C^1(\mathbb{R}^d, \mathbb{R}), f \in C^1(\mathbb{R}^d, \mathbb{R}^d), g \in C^1(\mathbb{R}^d, \mathbb{R}^{d\times k})$. We assume the minimizer exists.

The first method to solve the optimization problem involves computing a state-adjoint-control triple originating from the initial condition x, known as the open-loop control. The Pontryagin's Maximum Principle(PMP) yields the first-order optimality condition in the form of a forward-backward coupling

Hamilton Jacobi ODE:

$$\text{(TPBVP)} \quad \begin{cases} \dfrac{d}{dt}y^* = f(y^*) + g(y^*)u^*, \\ y^*(t_0) = x, \\ -\dfrac{d}{dt}p^* = \partial_y\Big(f(y^*) + g(y^*)u^*\Big)^T p^* + \partial_y l(y^*), \\ p^*(T) = 0. \end{cases}$$

By the maximality condition of the Hamiltonian at the optimal control we get the optimality condition

$$(1) \qquad u^* = -\frac{1}{2\beta}g(y^*)^T p^* \quad \forall t \in (t_0, T).$$

The optimal trajectories obtained from the solution of the open-loop system are not robust with respect to disturbance [1], since the existing numerical methods for two-point boundary value problem in general tend to diverge. The closed-loop method, on the other hand, is based on the dynamic programming. By introducing the value function

$$V(t, y) := \inf_u J(u; t, y) \quad \text{subject to the dynamic system,}$$

we synthesise the control by the maximum principle:

$$u^*(t, x) = -\frac{1}{2\beta}g(y^*)^T p^*,$$

and get a nonlinear hyperbolic partial differential equation, the Hamilton-Jacobi-Bellman (HJB) equation:

$$\text{(HJB)} \quad \begin{cases} \partial_t V - \dfrac{1}{4\beta}\nabla V(t,x)^T g(x) g^T(x) \nabla V(t,x) + \nabla V(t,x)^T f(x) + l(x) = 0 \\ V(T, x) = 0. \end{cases}$$

In this case the optimal control problem becomes solving a nonlinear hyperbolic partial differential equation. The method is more robust compared to the open-loop method but suffers from 2 main drawbacks. First, in the case where the dimension of the state space is large, we encounter the *curse of dimensionality(Bellman)* and the computational cost is formidable if it is not impossible. Second, since we usually linearise the system locally and solve the Riccati equation, the optimal control differs from the Riccati-based approximation especially in the transient phase. Many existing methods are provided to overcome the curse of dimensionality, in particular, the data-driven approaches. A data-driven approach usually approximates the optimal value function on a causality-free data set of state-value pair. The model does not require a grid and can be applied to high dimensional problem. In particular, there is a growing literature on solving the high-dimensional problem using neural network.

2.1. **theoretical review and notations.** Throughout the paper we denote Ω as $\mathbb{R}^{d+1}$, $M(\Omega)$ the space of Radon measure on Ω, $C_0(\Omega)$ the space of continuous functions that vanish at infinity. Without further specification, we denote $\langle \cdot, \cdot \rangle$ the dual bracket $\langle \cdot, \cdot \rangle_{M(\Omega), C(\Omega)}$.

We introduce the following notation:

- $M^0(\Omega)$: the space of all nonatomic measures on (Ω).
- $M^\sharp(\Omega)$: the space of all purely atomic measures on Ω.

We decompose $\mu \in M(\Omega)$ by $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$, where $\mu_{\text{cont}} \in M^0(\Omega)$ has support $\operatorname{cont} \mu$ and $\mu_{\text{atom}} \in M^\sharp(\Omega)$ has support $\operatorname{atom} \mu$. Therefore we have

$$M(\Omega) = M^0(\Omega) \oplus M^\sharp(\Omega).$$

3. THE SPARSE LEARNING PROBLEM

3.1. **the integral representation.** We use the shallow neural network to approximate the value function because its theoretical property is well-studied. The nice properties of the integral representation of the neural network has been extensively investigated. In [3] and [10] the convex nature of the shallow neural network has been proved. In [11] and [13] the representability of the shallow ReLU neural network is defined, the functional class, which can be represented by the integral neural network with finite Radon measure, is specified. We introduce the following notation: Ω a subset of $\mathbb{R}^{n+1}$, $D \subset \mathbb{R}^d$, $M(\Omega)$ the measure on Ω with bounded variation. For $\mu \in M(\Omega)$, we define the operator $\mathcal{N} : M(\Omega) \to L^2(D, \nu)$:

$$[\mathcal{N}\mu](x) = \int_\Omega \sigma(\omega, x) d\mu(\omega).$$

Since for all x, $\sigma(\cdot, x)$ is bounded by a μ-integrable function , and for all ω, $\sigma(\omega, \cdot)$ is continuous, we can differentiate under the integral sign and get

$$\nabla_x [\mathcal{N}\mu](x) = \int_\Omega \nabla_x \sigma(\omega, x) d\mu(\omega).$$

To abuse the notation, we define the operator

$$\nabla\mathcal{N} : M(\Omega) \to L^2(D, \nu)^d, \quad [\nabla\mathcal{N}\mu](x) = \nabla_x [\mathcal{N}\mu](x).$$

In the case that the measure is supported on finite number of atoms, it can be written as

$$[\mathcal{N}\mu](x) = \sum_{n=1}^N c_n \sigma(a_n \cdot x + b_n) = \sum_{n=1}^N c_n \operatorname{ReLU}(a_n \cdot x + b_n)^p,$$

which is the representation of the neural network, and the gradient of the model is:

$$[\partial_{x_i} \mathcal{N}](x) = \sum_{n=1}^N c_n \partial_{x_i} \sigma(a_n \cdot x + b_n) = \sum_{n=1}^N c_n p a_{n,i} \operatorname{ReLU}(a_n \cdot x + b_n)^{p-1}$$

3.2. **the activation functions.** After introducing the general problem, we formulate the concrete optimization problem. More specifically, we give the choice of activation functions and the sparsity-induced penalty term. First we consider the activation functions.

3.2.1. *general assumptions.*

Assumption 3.1 (on activation function)**.**

(1) σ is differentiable w.r.t. x.
(2) There exists $\Lambda > 0$, such that for all $x \in D$, $\omega_1, \omega_2 \in \Omega$, it holds

$$|\sigma(x,\omega_1) - \sigma(x,\omega_2)| \leq \Lambda(1+\|x\|)|\omega_1 - \omega_2|,$$
$$|\partial_{x_i}\sigma(x,\omega_1) - \partial_{x_i}\sigma(x,\omega_2)| \leq \Lambda(1+\|x\|)|\omega_1 - \omega_2| \text{ for all } i = 1,\dots,d.$$

The Lipschitz continuity of each partial derivative is also required since we are including the gradient information of the value function. In [1] the author explored the relationship between (TPBVP) and (HJB) and suggested a gradient-augmented approach. In the data generating process, the adjoint state p^* is obtained as a byproduct of solving (TPBVP). Moreover, the adjoint state provides the gradient information of the optimal value function: $p^*(t) = \nabla V(t, y^*(t))$ ([7] [5] [6]). Instead of state-value pair we get the dataset of the triple $\{x^i, V(x^i), \nabla V(x^i)\}_{i=1}^K$ that is enriched with gradient information. In the setting of supervised learning we use the neural network to approximate the optimal value function:

$$\mathcal{N}_{\omega,c}(x) = \sum_{n=1}^{N} c_n \sigma(a_n \cdot x + b_n) \in \mathbb{R}, \tag{2}$$

where σ is a sufficiently smooth activation function and N the width of the network. And the gradient is

$$\nabla_x \mathcal{N}_{\omega,c}(x) = \sum_{n=1}^{N} c_n \nabla_x \sigma(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

We have the minimization problem

$$\min_{N\in\mathbb{N},(a_n,b_n,c_n)\in\mathbb{R}^d\times\mathbb{R}\times\mathbb{R},} l_K(\mathcal{N}_{\omega,c}; V),$$

with

$$l_K(\mathcal{N}_{\omega,c}; V) := \frac{1}{2K}\left(\sum_{k=1}^{K} |V(x^k) - \mathcal{N}_{\omega,c}(x^k)|^2 + \sum_{k=1}^{K}\sum_{i=1}^{d} |\partial_{x_i} V(x^k) - \partial_{x_i}\mathcal{N}_{\omega,c}(x^k)|^2\right). \tag{3}$$

For large datasets we may interpret x^k to be random sample of the random variable $x \in L^2(D,\nu)$ with probability distribution ν on the domain D. The loss function can be understood as the empirical estimation of the loss

$$l_\nu(\mathcal{N}_{\omega,c}; V) = \frac{1}{2}\int_D |V(x) - \mathcal{N}_{\omega,c}(x)|^2 + \sum_{i=1}^{d} |\partial_{x_i} V(x) - \partial_{x_i}\mathcal{N}_{\omega,c}(x)|^2 d\nu.$$

3.2.2. *the p-homogeneous activation function.*

$$\sigma(x,\omega) = \max\{0, \omega \cdot x\}^p, \text{ with } p \in (2,\infty).$$

In this case the gradient is

$$\nabla_x \sigma(x,\omega) = \begin{bmatrix} p\max\{0,\omega\cdot x\}^{p-1}\omega_1 \\ p\max\{0,\omega\cdot x\}^{p-1}\omega_2 \\ \cdots \\ p\max\{0,\omega\cdot x\}^{p-1}\omega_d \end{bmatrix} \in \mathbb{R}^d.$$

The activation functions are non-decreasing and positive homogeneous of degree p, and it admits the following nice properties.

- Since they are not polynomial, they can linearly approximate any measurable function [9].
- They are invariant w.r.t. the change of scale of data, which we will explain further in detail.

In the case where $p = 1$ we have the ReLU functions, which has been extensively studied recently. For $p > 1$, the activation function not only preserves the nice property such as homogeneity of ReLU, but also smooths the kinks. We gain 2 things from the differentiability: 1. It circumvents the non-differentiable corners of the ReLU function; 2. The gradient information of the activation function can be used to calculate the gradient-augmented loss function. More generally, we assume the activation function to satisfy the following condition:

Assumption 3.2 (p-homogeneity)**.** σ is p-homogeneous w.r.t. ω for some $p > 2$, i.e., for all $x \in D$, $\lambda > 0$ it holds

$$\sigma(x, \lambda\omega) = \lambda^p \sigma(x,\omega) \quad \text{for all } \omega \in \mathbb{R}^{d+1}.$$

Remark 1. For $p > 2$, the activation function is twice differentiable and the gradient-augmented optimization problem is a smooth optimization problem, the value part of the loss is approximated by a high-order ReLU activation function $\mathrm{ReLU}^{2+\epsilon}$ and gradient is approximated by $\mathrm{ReLU}^{1+\epsilon}$ for a small ϵ. In the case of $p = 2$, the optimization problem is still well defined but we would have non-differentiable corners for $\partial_{x_i}\sigma$ as in the ReLU case. For $p < 2$, the Lipschitz continuity of the gradient function is compromised.

Define the neural network with N neurons:

$$\mathcal{N}_{\omega,c} : \mathbb{R}^d \to \mathbb{R}, \quad x \mapsto \sum_{n=1}^{N} c_n \sigma(a_n \cdot x + b_n), \quad c_n, b_n \in \mathbb{R}, a_n \in \mathbb{R}^d.$$

First, we observe the optimization problem with L^2-constrained inner weights and nonconvex-penalized outer weights. We prove that in this setting the nice property of p-homogeneous active function still holds: by proportionally linearizing the outer weights we can constrain the inner weights on a unit sphere rather than a unit ball.

For enforcing the sparsity we implement the explicit sparsity-promoting penalty term. Recently, there is the thought-provoking result in [14] that indicates normal explicit generalization methods including weight decay do not enhance the generalization performance for large neural network. Nevertheless, the explicit regularization still plays a central role in the sparse learning. The network training considered here is based on the following optimization problem:

$$\min_{N\in\mathbb{N},\|a_n\|^2+|b_n|^2\leq 1, c_n\in\mathbb{R}, n=1,\dots,N.} l_K(\mathcal{N}_{\omega,c}(x);V)+\alpha\sum_{n=1}^{N}\phi(|c_n|), \tag{4}$$

where the penalty term ϕ is observed under the following Assumption 3.3, 3.4 .

The following proposition states that, any optimal solution of (4) can be contracted on the unit sphere without increasing its value of the cost function. Hence we may constrain the optimization problem to a unit sphere.

Proposition 3.1 (homogeneous reformulation)**.** *Assuming (4) admits at least 1 global optimal solution and σ is p-homogeneous with $p\geq 1$, then (4) is equivalent to*

$$\min_{N\in\mathbb{N},\|a_n\|^2+|b_n|^2=1, c_n\in\mathbb{R}, n=1,\dots,N.} l_K(\mathcal{N}_{\omega,c}(x);V)+\alpha\sum_{n=1}^{N}\phi(|c_n|). \tag{P_ϕ}$$

Proof. By assumption the optimal solution exits, denote $(\hat{N},(\hat{a}_n,\hat{b}_n,\hat{c}_n)_{n=1}^{\hat{N}})$ one of the global minimizer. The optimal value of the first part in the fidelity term is

$$\frac{1}{2K}\sum_{k=1}^{K}\left|\left(\sum_{n=1}^{\hat{N}}\hat{c}_n\sigma(\hat{a}_n\cdot x_k+\hat{b}_n)\right)-V(x_k)\right|^2.$$

For $(\hat{a}_n,\hat{b}_n)$ in the interior of the unit ball with $\|\hat{a}_n\|^2+|\hat{b}_n|^2\lneqq 1$, denote $\lambda_n=\frac{1}{\sqrt{\|\hat{a}_n\|^2+|\hat{b}_n|^2}}\gneqq 1$. Projecting $(\hat{a}_n,\hat{b}_n)$ on the sphere by defining

$$(\tilde{a}_n,\tilde{b}_n)=\lambda_n(\hat{a}_n,\hat{b}_n).$$

Rescaling the outer weights by define $\tilde{c}_n=\frac{1}{\lambda_n^p}\hat{c}_n$, by p-homogeneity of the activation the function value stays the same:

$$\begin{aligned}\frac{1}{2K}\sum_{k=1}^{K}\left|\left(\sum_{n=1}^{\hat{N}}\tilde{c}_n\sigma(\tilde{a}_n\cdot x_k+\tilde{b}_n)\right)-V(x_k)\right|^2&=\frac{1}{2K}\sum_{k=1}^{K}\left|\left(\sum_{n=1}^{\hat{N}}\hat{c}_n\frac{1}{\lambda_n^p}\lambda_n^p\sigma(\hat{a}_n\cdot x_k+\hat{b}_n)\right)-V(x_k)\right|^2,\\&=\frac{1}{2K}\sum_{k=1}^{K}\left|\left(\sum_{n=1}^{\hat{N}}\hat{c}_n\sigma(\hat{a}_n\cdot x_k+\hat{b}_n)\right)-V(x_k)\right|^2.\end{aligned}$$

The fidelity is invariant w.r.t. the scaling. The similar result holds for the second part of the fidelity term by linearity

$$\tilde{c}_n\nabla_x\sigma(\tilde{a}_n\cdot x_k+\tilde{b}_n)=\hat{c}_n\frac{1}{\lambda_n^p}\nabla_x\left(\lambda_n^p\sigma(\hat{a}_n\cdot x+\hat{b}_n)\right)=\hat{c}_n\nabla_x\sigma(\hat{a}_n\cdot x+\hat{b}_n).$$

On the other hand, the penalty term is nondecreasing by Assumption 3.3. Since $\lambda_n > 1$ and $p \geq 1$, it holds $|\tilde{c}_n| = |\hat{c}_n|/\lambda_n^p \leq |\hat{c}_n|$, hence

$$\phi(|\tilde{c}_n|) \leq \phi(|\hat{c}_n|),$$

The pair $\left(\tilde{N}, (\tilde{a}_n, \tilde{b}_n, \tilde{c}_n)\right)_{n=1}^{\tilde{N}}$ with

$$\tilde{N} = \hat{N} \in \mathbb{N}, \quad (\tilde{a}_n, \tilde{b}_n) \in \mathbb{S}^d, \quad \tilde{c}_n \in \mathbb{R}, \quad n = 1, \ldots, \tilde{N},$$

admits smaller value. The statement holds. □

The proof relies essentially on the homogeneity of the activation function and the monotonicity of the penalty functions.

3.3. **the nonconvex sparse learning approach.** As introduced in the abstract, the l^1 penalty fails to detect the redundancy of the neurons clustering in a small neighborhood, which motivates the nonconvex penalty term. First we introduce the following assumption:

Assumption 3.3. The penalty term $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ satisfies:

(1) ϕ is differentiable at 0 with $\phi(0) = 0$ and $\phi'(0) = 1$.
(2) ϕ is differentiable, concave, nondecreasing and coercive($\phi(z) \to \infty$ as $z \to \infty$).

The Assumption 3.3 describe the global property of the penalty functions. By introducing the nonconvexity the coercivity of the quadratic cost is compromised. With Assumption 3.3 the optimization problem stays coercive such that we can apply the direct method of calculus of variation. The Assumption 3.3 (1) depends on the choice of activation functions and may be adjusted via linear transformation as long as the derivative at the given point stays bounded. The following lemma follows from the assumption.

Lemma 3.2. *If Assumption 3.3 is satisfied, then it holds*

(1) *subadditivity: for all a, b, $\phi(a + b) \leq \phi(a) + \phi(b)$.*
(2) $\phi(|z|) \leq |z|$.

Proof.
1. It holds

$$\begin{aligned} \phi(z_1 + z_2) &= \int_0^{z_1} \phi'(z)dz + \int_0^{z_2} \phi'(z_1 + z)dz, \\ \text{(concavity)} \quad &\leq \int_0^{z_1} \phi'(z)dz + \int_0^{z_2} \phi'(z)dz, \\ &= \phi(z_1) + \phi(z_2). \end{aligned}$$

2. it holds for $z \geq 0$:

$$\begin{aligned} \phi(z_1) &= \int_0^{z_1} \phi'(z)dz \\ (\phi \text{ concave and } \phi'(0) = 1) &\leq \int_0^{z_1} 1dz \\ &= z_1 \end{aligned}$$

□

Assumption 3.3 serves the purpose for the nonconvex problem to admits a global minimizer. To enhance the sparsity, we need to introduce other assumption. In particular, It needs to exert strong penalizing effect on the weight on 2 neighboring neurons such that merging them would in fact reduce the cost. To this purpose, we need the strict concavity to enforce strict subadditivity, and we only need that in a neighborhood of 0:

Assumption 3.4 (Local concavity)**.** The penalty term $\phi : \mathbb{R}_+ \to \mathbb{R}$ satisfies:

(1) There exists $\gamma_1 \geq 0, \hat{z}_1 > 0$, such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for all $0 \leq z_1 \leq z_2 \leq \hat{z}_1$. ($\gamma$-convex)

(2) There exists $\gamma_2 > 0$, $\hat{z}_2 > 0$, such that $\phi'(z_2) - \phi'(z_1) \leq -\gamma_2(z_2 - z_1) \leq 0$ for all $0 \leq z_1 \leq z_2 \leq \hat{z}_2$. (strongly concave)

Remark 2. The Assumption 3.4 can be interpreted as the concave function being strongly concave but exhibits also certain convexity locally at 0. It is worth noticing that it is entirely a local property that only needs to take effect at 0. Assumption 3.4(2) induces strict subadditivity that takes effect at small weights.

In general the global concavity in the Assumption 3.3 doesn't imply locally uniform concavity in the Assumption 3.4(2). Observe, for example, $\phi(x) = x - \frac{2}{3}x^{\frac{3}{2}}$, where the function satisfies the Assumption 3.3 but the locally uniform concavity fails. However since Assumption 3.4 is only local property we may construct piecewise penalty by first constructing the penalty around the neighborhood of 0.

Proposition 3.3 (the second order estimate)**.** *Given ϕ that satisfies the Assumption 3.3, if ϕ satisfies further Assumption 3.4, then it holds*

(1) *for all* $z \in [0, \hat{z}_1] : \phi(z) \geq z - \frac{\gamma_1}{2}z^2$,

(2) *for all* $z \in [0, \hat{z}_2] : \phi(z) \leq z - \frac{\gamma_2}{2}z^2$.

By the proposition above we know any penalty satisfies Assumption 3.3 and 3.4 admits such upper and lower bound. The function

$$\phi_{\log,\gamma}(z) = \frac{1}{\gamma}\log(1 + \gamma z) \tag{5}$$

for $\gamma > 0$ and its convex combination with the $l1$ norm will be the main function of choice for our numerical analysis.

Next we need to incorporate the nonconvex penalty in the integral representation. For the atom parts it is defined in the discrete setting. For the continuous part on Ω we assume no nonconvex penalty and the measure is penalized with the usual l_1 norm. Concretely, observe $\Omega = \operatorname{atom}\mu \cup \operatorname{cont}\mu$, where $\operatorname{atom}\mu$ and $\operatorname{cont}\mu$ are defined by the Lebesgue-Nikodym decomposition, we define the penalty term

$$\Phi(\mu) = \|\mu_{\text{cont}}\|_{M(\Omega)} + \sum_{\omega \in \operatorname{atom}\mu} \phi(|\mu|(\omega)). \tag{6}$$

We formulate now the optimization problem as following:

$$(P_\Phi) \quad \begin{cases} \min\limits_{\mu \in M(\Omega)} J(\mu) := L(\mu) + \alpha\Phi(\mu), \\ \text{with,} \\ L(\mu) = \dfrac{1}{2}\left(\|[\mathcal{N}\mu] - V\|^2_{L^2(D,\nu)} + \sum\limits_{i=1}^{d} \|\partial_{x_i}[\mathcal{N}\mu] - \partial_{x_i} V\|^2_{L^2(D,\nu)} \right). \end{cases}$$

4. EXISTENCE OF MINIMIZERS

4.1. **existence of the global minimizer.** In this subsection we prove the existence of the global minimizer. We leverage the direct method of calculus variation. Since we are working on the space of measure $M(\Omega)$ we consider the weak -* convergence. As usual, the common difficulty is to prove the weak-* l.s.c.. Furthermore, the coercivity is compromised by introducing the nonconvex penalty term Φ. The next lemma address this problem. It proves that the coercivity of Φ is preserved by the Assumption 3.3 of ϕ.

Lemma 4.1 (On the coercivity of Φ)**.** *Under the Assumption 3.3, for any $\mu \in M(\Omega)$, it holds*

$$\phi(\|\mu\|_{M(\Omega)}) \leq \Phi(\mu) \leq \|\mu\|_{M(\Omega)}.$$

In particular, the functional Φ is coercive.

Proof. The proof relies on the subadditivity and $\phi(|z|) \leq |z|$.

$$\begin{aligned} \phi(\|\mu\|_{M(\Omega)}) &= \phi\left(|\mu|(\Omega \setminus \operatorname{atom}\mu) + \sum_{\omega \in \operatorname{atom}\mu} |\mu|(\{\omega\}) \right), \\ \text{(subadditivity)} \quad &\leq \phi(|\mu|(\Omega \setminus \operatorname{atom}\mu) + \sum_{\omega \in \operatorname{atom}\mu} \phi(|\mu|(\{\omega\})), \\ \left(\phi(|z|) \leq |z|\right) \quad &\leq |\mu|(\Omega \setminus \operatorname{atom}\mu) + \sum_{\omega \in \operatorname{atom}\mu} \phi(|\mu|(\{\omega\})), \\ &= \Phi(\mu). \end{aligned}$$

In particular, by the coercivity of ϕ it holds that Φ is also coercive.
For the second inequality, we leverage that ϕ is sublinear and $\phi(0) = 0$. We get $\Phi(\mu) \leq \|\mu\|_{M(\Omega)}$. □

The next lemma proves the weak-* l.s.c. of Φ. The proof follows the idea in the paper [4]. Since the original proof is valid for Ω locally compact, the result applies to the current setting. Moreover, since our functional doesn't have the singular part, the proof and assumptions can be simplified.

Lemma 4.2. *Given $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$, under Assumption 3.3, it holds*

$$\Phi(\mu) = \int_\Omega d\mu_{\text{cont}} + \sum_{\omega \in \text{atom}\, \mu} \phi(|\mu_{\text{atom}}|(\{\omega\}))$$

is weak- sequentially l.s.c..*

Proof. By proposition 2.5 in [4] it is equivalent to show the weak-* l.s.c. of $\Phi_\epsilon(\mu) := \Phi(\mu) + c\|\mu\|_{M(\Omega)}$ for each $c > 0$. We can therefore assume there exists $c > 0$, such that

$$\Phi(\mu) \geq c|\mu|(\Omega) \quad \forall \mu \in M(\Omega).$$

Let $\mu_h \overset{*}{\rightharpoonup} \mu$, we decompose Ω by heavy atomes and small atoms combined the continuous part:

$$\Omega = A_h + (\Omega - A_h), \text{ where } A_h := \{\omega \in \Omega : \mu_h(\omega) \geq \epsilon\}.$$

By passing to a subsequence we have

$$\mu_{h,\text{cont}} \overset{*}{\rightharpoonup} \mu_{\text{cont}}, \quad \mu_{h,\text{atom}} \overset{*}{\rightharpoonup} \mu_{\text{atom}}$$

We prove the weak-* l.s.c. of each part.

On A_h:

Since μ_h has bounded variation and $\mu_h(\omega) \geq \epsilon$ for $\omega \in A_h$, we know the cardinality of A_h is bounded. By lemma 3.6 in [4] we know $M(A_h)$ is sequentially weak-* closed. Moreover, $\Phi(\mu_h \mathbb{1}_{A_h})$ is weak-* l.s.c. if ϕ is l.s.c. and subadditive, therefore we have

$$\liminf_h \Phi(\mu_h \mathbb{1}_{A_h}) = \liminf \sum_{\omega \in \text{atom}\, \mu_h \cap A_h} \phi(|\mu_{h,\text{atom}}|(\{\omega\})) \geq \Phi(\mu \mathbb{1}_{A_h}).$$

On $\Omega - A_h$:

By proposition 3.3, we have for ϵ small enough, for all $z \in [0, \epsilon]$:

$$\phi(z) \geq z - \frac{\gamma_1 z^2}{2},$$
$$\phi(z) \leq z.$$

We have

$$\begin{aligned}
&\liminf_h \Phi(\mu_h \mathbb{1}_{\Omega - A_h}) \\
&= \liminf_h \left(\int_{\Omega - A_h} d\mu_{\text{cont}} + \sum_{\omega \in (\Omega - A_h) \cap \text{atom}\, \mu_h} \phi(|\mu_{h,\text{atom}}|(\{\omega\})) \right) \\
&\geq \liminf_h \left(\int_{\Omega - A_h} d\mu_{\text{cont}} + \sum_{\omega \in (\Omega - A_h) \cap \text{atom}\, \mu_h} |\mu_{h,\text{atom}}|(\{\omega\}) - \frac{\gamma_1}{2} |\mu_{h,\text{atom}}|(\{\omega\})^2 \right).
\end{aligned}$$

Choose $\delta \geq \frac{\gamma_1}{2}\epsilon$, we have

$$\sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} |\mu_{h,\text{atom}}|(\{\omega\}) - \frac{\gamma_1}{2}|\mu_{h,\text{atom}}|(\{\omega\})^2 \geq \sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} (1-\delta)|\mu_{h,\text{atom}}|(\{\omega\}).$$

Hence

$$\begin{aligned}
&\liminf_h \Phi(\mu_h \mathbb{1}_{\Omega-A_h}) \\
\geq &\liminf_h (1-\delta)\int_{\Omega-A_h} d\mu_{\text{cont}} + \liminf_h (1-\delta) \sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} |\mu_{h,\text{atom}}|(\{\omega\}) \\
= &\liminf_h (1-\delta)\|\mu_h \mathbb{1}_{\Omega-A_h}\|_{M(\Omega)}.
\end{aligned}$$

By the weak-* l.s.c. of the norm $\|\cdot\|_{M(\Omega)}$ we have

$$\liminf_h \Phi(\mu_h \mathbb{1}_{\Omega-A_h}) \geq (1-\delta)\|\mu \mathbb{1}_{\Omega-A_h}\|_{M(\Omega)}.$$

Furthermore, applying $\phi(z) \leq z$ we get

$$\sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} |\mu_{\text{atom}}|(\{\omega\}) \geq \sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} \phi(|\mu_{h,\text{atom}}|(\{\omega\})).$$

Therefore

$$\begin{aligned}
&\liminf_h \Phi(\mu_h \mathbb{1}_{\Omega-A_h}) \\
\geq &(1-\delta)\|\mu \mathbb{1}_{\Omega-A_h}\|_{M(\Omega)} \\
\geq &(1-\delta)\int_{\Omega-A_h} d\mu_{\text{cont}} + \sum_{\omega\in(\Omega-A_h)\cap\text{atom}\,\mu_h} \phi(|\mu_{h,\text{atom}}|(\{\omega\})) \\
= &(1-\delta)\Phi(\mu \mathbb{1}_{\Omega-A_h}).
\end{aligned}$$

Taking $\epsilon \to 0$ we get $\delta \to 0$ and the result is proven. □

Theorem 4.3 (Existence of the global minimizer)**.** *Assuming Assumption 3.3 is fulfilled, problem (P_Φ) admits at least one global minimizer in $M(\Omega)$.*

Proof. We apply the direct method of the calculus of variation:
1. Boundedness (ability to take a minimizing sequence in $M(\Omega)$): $J \geq 0$, therefore J is bounded from below.
2. Minimizing sequence admits a weak-* convergent subsequence: To apply Banach-Alaoglu we need to prove that every minimizing sequence of J is bounded in $M(\Omega)$. We prove it by the usual coercivity argument. By the non-convexity of the penalty term, the coercivity of the cost function is compromised. However, we get form lemma 4.1

$$\alpha\|\mu\|_{M(\Omega)} \leq \alpha\Phi(\mu) \leq J(\mu),$$

hence J is also coercive.
3. Lower-semi continuity: by the lemma $\Phi(\mu)$ is weak-* l.s.c., therefore we have for $\mu_n \overset{*}{\rightharpoonup} \hat{\mu}$,

$$\begin{aligned}\liminf_{n\to\infty} J(\mu_n) &= \liminf_{n\to\infty}\Big(L(\mu_n)+\alpha\Phi(\mu_n)\Big)\\ &\geq L(\hat{\mu})+\alpha\liminf_{n\to\infty}\Phi(\mu_n)\\ &\geq L(\hat{\mu})+\alpha\Phi(\hat{\mu})\\ &= J(\hat{\mu}),\end{aligned}$$

where the inequality in the second line follows from the weak-* l.s.c. of L (since L is convex and continuous composed with the weak-*-to-weak continuous operator $\mathcal{N}$) and the inequality in the third line follows from weak-* l.s.c. of the functional Φ. □

4.2. **optimality condition and finite support of the solution.** Since the optimization problem is nonconvex, the global minimizer is difficult to find in general. Instead, we define the local minimum:

Definition 4.1 (local minimizer)**.** $\bar{\mu} \in M(\Omega)$ is a local minimizer if there exists $\epsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \quad \text{for all } \mu \text{ with } \|\mu-\bar{\mu}\|_{M(\Omega)} \leq \epsilon.$$

Remark 3 (The gradient of J)**.** To further investigate the optimality condition of (P_Φ) and derive the according algorithm, we calculate the first variation of $J(\mu)$, We define the gradient information $\bar{p}(\omega) \in C(\Omega)$ as

$$\begin{aligned}\bar{p}(\omega) &:= \nabla L(\bar{\mu})\\ &= \mathcal{N}^*(\mathcal{N}[\bar{\mu}]-V)+\sum_{i=1}^{d}(\partial_{x_i}\mathcal{N})^*(\partial_{x_i}\mathcal{N}[\bar{\mu}]-\partial_{x_i}V)\\ &= \int_D \sigma(x,\omega)(\mathcal{N}[\bar{\mu}](x)-V)d\nu(x)+\sum_{i=1}^{d}\int_D \partial_{x_i}\sigma(x,\omega)(\partial_{x_i}\mathcal{N}[\bar{\mu}](x)-\partial_{x_i}V)d\nu(x).\end{aligned} \tag{7}$$

For the finite sample and $\sigma(x) = \mathrm{ReLU}(x)^p$, the gradient admits the form:

$$p(\omega) = \frac{1}{K}\sum_{k=1}^{K}\mathrm{ReLU}(\omega\cdot x_k)^p(\mathcal{N}_{\bar{\mu}}(x_k)-y_k)+\frac{1}{K}\sum_{k=1}^{K}\sum_{i=1}^{d}p\cdot\mathrm{ReLU}(\omega\cdot x_k)^{p-1}\omega_i(\partial_{x_i}\mathcal{N}(x_k)-\partial_{x_i}V(x_k)) \tag{8}$$

Theorem 4.4 (the optimality condition)**.** *Under the Assumption 3.3, if $\bar{\mu}$ is a local minimum of J, then it holds*

(1) $|\bar{p}| \leq \alpha$ *for all* ω.
(2) $\bar{p}(\omega) = -\alpha\phi'(|\bar{\mu}|(\omega))\,\mathrm{sign}(\bar{\mu}\{w\})$, *for* $\bar{\mu}$*-almost all* $\omega \in \Omega$.

the signum function $\mathrm{sign}: \Omega \to \{-1,1\}$ *is defined* μ*-a.s. by Hahn decomposition.*

Proof. We leverage that dual evaluation of δ_ω is the evaluation of $\bar{p}$ at ω: $\langle \bar{p}, \delta_\omega \rangle_{C(\Omega),M(\Omega)} = \bar{p}(\omega)$. Since $\bar{\mu}$ is the local minimizer, it holds for all $u \in M(\Omega)$:

$$J(\bar{\mu} + \tau u) - J(\bar{\mu}) = L(\bar{\mu} + \tau u) - L(\bar{\mu}) + \alpha\Phi(\bar{\mu} + \tau u) - \alpha\Phi(\bar{\mu}) \geq 0.$$

Taking $\tau \to 0$, it holds

$$-\langle \bar{p}, u \rangle \leq \alpha \lim_{\tau \to 0} \frac{1}{\tau}(\Phi(\bar{\mu} + \tau u) - \Phi(\bar{\mu})).$$

Now we separate 2 cases:

1) For the atom part, $\omega \in \operatorname{atom} \bar{\mu}$ with $\bar{\mu}(\{\omega\}) = c$, taking $u = \delta_\omega$. it holds

$$-p(\omega) = -\langle \bar{p}, u \rangle \leq \alpha \lim_{\tau \to 0} \frac{1}{\tau}(\phi(|c + \tau|) - \phi(|c|) = \alpha\phi'(|c|) \operatorname{sign}(c).$$

Taking $u = \pm\delta_\omega$, we get $p(\omega) = -\alpha\phi'(|c|) \operatorname{sign}(c)$. Since $\phi'(0) = 1$ and ϕ' is decreasing we have $\phi'(|c|) \in [0, 1]$, hence $|\bar{p}| \leq \alpha$.

2) For the continuous part, first observe $\omega \notin \operatorname{atom} \bar{\mu}$ and perturb with $u = \delta_\omega$, we have

$$-\bar{p}(\omega) = -\langle p, u \rangle \leq \alpha \lim_{\tau \to 0} \frac{1}{\tau}(\phi(\tau) - \phi(0)) = \alpha\phi'(0) = \alpha.$$

Taking $u = -\delta_\omega$ we get $|p(\omega)| \leq \alpha$ for all $\omega \notin \operatorname{atom} \bar{\mu}$. Next we take $u = \bar{\mu}_{\text{cont}}$, we have

$$-\langle \bar{p}, \bar{\mu}_{\text{cont}} \rangle \leq \alpha \lim_{\tau \to 0} \frac{1}{\tau}\tau\|\bar{\mu}_{\text{cont}}\|_{M(\Omega)} = \alpha\|\bar{\mu}_{\text{cont}}\|_{M(\Omega)}.$$

Taking $u = -\bar{\mu}_{\text{cont}}$ we get $-\langle \bar{p}, \bar{\mu}_{\text{cont}} \rangle = \alpha\|\bar{\mu}_{\text{cont}}\|_{M(\Omega)}$, i.e.,

$$\int_\Omega \Big(-\bar{p}(\omega) * \operatorname{sign} \bar{\mu}_{\text{cont}}(\omega) - \alpha \Big) d|\bar{\mu}|(\omega) = 0.$$

By $|p(\omega)| \leq \alpha$ we get $\bar{p} = -\alpha * \operatorname{sign} \bar{\mu}_{\text{cont}}$ for almost all $\omega \in \operatorname{cont} \bar{\mu}$. □

If the sample size is finite, then the optimal solution admits finite support by the Carathéodory's theorem. The following theorem proves that the optimal solution admits the finite support even though the dataset is infinitely large.

Theorem 4.5. *If the Assumption 3.3 and the Assumption 3.4 are satisfied, then the optimal solution of (P_Φ) admits at most finite nodes.*

Proof.

1. $\operatorname{cont} \bar{\mu} = \emptyset$: Argue by contradiction. Assuming there exists $\hat{\omega}$ such that $\bar{\mu}(\{\hat{\omega}\}) = 0$ and $\delta > 0$ such that the open ball $B_\delta(\hat{\omega})$ on Ω satisfies $|\bar{\mu}|(B_\delta(\hat{\omega})) > 0$. Define $D_\delta := B_\delta(\hat{\omega})$ and $C_\delta := |\bar{\mu}|(D_\delta)$ and construct the measure

$$\mu_\delta = \bar{\mu} - \bar{\mu}|_{D_\delta} + C_\delta \delta_{\hat{\omega}}.$$

The cost function is quadratic, therefore the second order Taylor expansion is exact, we get:

$$L(\mu_\delta) = L(\bar{\mu}) + \langle \bar{p}, \mu_\delta - \bar{\mu} \rangle_{C(\Omega),M(\Omega)} + \frac{1}{2}\|\mathcal{N}[u_\delta - \bar{\mu}]\|^2_{L^2(D,\nu)} + \frac{1}{2}\sum_{i=1}^{d} \|\partial_{x_i}\mathcal{N}[\mu_\delta - \bar{\mu}]\|^2_{L^2(D,\nu)}$$

By Assumption 3.1, it holds

$$\begin{aligned}
|\mathcal{N}[\mu_\delta - \bar{\mu}](x)| &\leq \left| \int_{D_\delta} \sigma(x,\omega) d\bar{\mu} - C_\delta \sigma(x,\hat{\omega}) \right|, \\
&= \left| \int_{D_\delta} \sigma(x,\omega) - \sigma(x,\hat{\omega}) d\bar{\mu} \right|, \\
&\leq \delta\Lambda(1 + \|x\|)|\bar{\mu}|(D_\delta), \\
&= \delta\Lambda(1 + \|x\|)C_\delta.
\end{aligned}$$

Analogously it holds

$$|\partial_{x_i}\mathcal{N}[\mu_\delta - \bar{\mu}](x)| \leq \delta\Lambda(1 + \|x\|)C_\delta.$$

Define

$$\Lambda_1^2(D,\Lambda) = \int_D (1 + \|x\|)^2 d\nu \Lambda^2.$$

Hence

$$L(\mu_\delta) = L(\bar{\mu}) + \langle \bar{p}, \mu_\delta - \bar{\mu} \rangle + \frac{d+1}{2}\delta^2\Lambda_1^2 C_\delta^2.$$

By the Theorem 4.4 $\bar{p}$ is constant for $\omega \in \operatorname{cont}\bar{\mu}$, therefore the first order term vanishes, it holds

$$L(\mu_\delta) = L(\bar{\mu}) + \frac{d+1}{2}\delta^2\Lambda_1^2 C_\delta^2.$$

Concerning the cost term, it holds

$$\begin{aligned}
\Phi(\mu_\delta) &= \Phi(\bar{\mu}) - \bar{\mu}(D_\delta) + \phi(C_\delta), \\
&= \Phi(\bar{\mu}) + \phi(C_\delta) - \bar{\mu}(D_\delta), \\
&= \Phi(\bar{\mu}) + \int_0^{C_\delta} \phi'(\xi) - \phi'(0) d\xi, \\
\text{Assumption 3.4(2)} \quad &\leq \Phi(\bar{\mu}) - \frac{\gamma_2}{2} C_\delta^2.
\end{aligned}$$

Combining the 2 parts we get

$$J(\mu_\delta) \leq J(\bar{\mu}) - \left(\frac{\alpha\gamma_2 - (d+1)\delta^2\Lambda_1^2}{2} \right) C_\delta^2.$$

Choose $\delta^2 \leq \frac{2\alpha\gamma_2}{(d+1)\Lambda_1^2(D,\Lambda)}$ we get the contradiction.

2. $|\operatorname{atom}\bar{\mu}|$ is finite: Without loss of generality we can assume the atomic support of μ is contained in a compact subset of Ω. Assume by contradiction, there exists a sequence $\omega_n \to \infty$ escapes to the infinity.

Since $\bar{\mu} \in M(\Omega)$ admits bounded variation, by passing to a subsequence, it holds $\bar{\mu}(\omega_n) \to 0$. Assuming w.l.o.g. that $\bar{\mu}(\omega_n) = c_n > 0$, we have

$$\lim_{\omega_n \to +\infty} \bar{p}(\omega_n) = -\alpha\phi'(|\bar{\mu}|(\omega_n)) \operatorname{sign}(\bar{\mu}(\omega_n)) = -\alpha,$$

which contradicts to that $\bar{p} \in C_0(\Omega)$ and $\lim_{\omega_n \to +\infty} \bar{p}(\omega_n) \to 0$.

Now we observe atom $\bar{\mu}$. Since it is contained in a compact subset of Ω, there exists a sequence ω_n (possibly passing through a subsequence) such that

$$\omega_n \to \hat{\omega} \text{ in } \Omega \quad \text{ for some } \hat{\omega} \in \Omega.$$

Assuming w.l.o.g. that $\bar{\mu}(\omega_n) = c_n > 0$, by the boundedness of total variation it holds $\lim_{n\to\infty} c_n = 0$. By the Theorem 4.4 , it holds

$$\bar{p}(\hat{\omega}) = \lim_{n\to\infty} \bar{p}(\omega_n) = -\alpha \lim_{n\to\infty} \phi'(c_n) \to -\alpha.$$

Concentrate the measure on $\{\omega_N, \dots\}$ on the limits $\hat{\omega}$, i.e., define

$$\mu_N := \bar{\mu} - \sum_{i=N}^{\infty} \delta_{\omega_i} \bar{\mu}(\omega_i) + C_N \delta_{\hat{\omega}}, \text{ where } C_N = \sum_{i=N}^{\infty} c_i > 0.$$

As in 1., the fidelity term of the cost can by decomposed by Taylor expansion.
Define $\delta_N := \max_{i=N}^{\infty} |\omega_i - \hat{\omega}|$:

$$L(\mu_N) \le L(\bar{\mu}) + \langle \bar{p}, \mu_N - \bar{\mu} \rangle + \frac{d+1}{2} \Lambda_1^2(D, \Lambda) \delta_N^2 C_N^2$$

For the first variation we have by the optimality condition and the argument above:

$$\langle \bar{p}, \mu_N - \bar{\mu} \rangle = \langle \bar{p}, -\sum_{i=N}^{\infty} \delta_{\omega_i} \bar{\mu}(\omega_i) + C_N \delta_{\hat{\omega}} \rangle = \alpha \sum_{i=N}^{\infty} c_i \phi'(c_i) - \alpha C_N.$$

For the penalty term, it holds

$$\Phi(\mu_N) = \Phi(\mu) - \sum_{i=N}^{\infty} \phi(c_i) + \phi(C_N).$$

Combining the estimation, it holds

$$J(\mu_N) \le J(\bar{\mu}) - \alpha \sum_{i=N}^{\infty} \Big(\phi(c_i) - c_i \phi'(c_i) \Big) - \alpha \Big(C_N - \phi(C_N) \Big) + \frac{d+1}{2} \Lambda_1^2(D, \Lambda) \delta_N^2 C_N^2.$$

By concavity of ϕ it holds

$$\phi(c_i) - c_i \phi'(c_i) \ge \phi(0) = 0.$$

By Assumption 3.4 (2) it holds

$$C_N - \phi(C_N) = C_N \phi'(0) - \Big(\phi(0) + \int_0^{C_N} \phi'(\xi) d\xi \Big),$$

$$\text{(Assumption 3.3)} = -\int_0^{C_N} \phi'(\xi) - \phi'(0) d\xi,$$

$$\geq \frac{\gamma_2}{2} C_N^2.$$

Thus

$$J(\mu_N) \leq J(\bar{\mu}) - \frac{1}{2}\Big(\alpha\gamma_2 - (d+1)\Lambda_1^2(D,\Lambda)\delta_N^2\Big)C_N^2.$$

Taking

$$\delta^2 < \frac{\alpha\gamma_2}{\Lambda_1^2(\Lambda, D)(d+1)},$$

we get the contradiction. □

Now we have a necessary optimality condition. But it is not sufficient. More importantly, we cannot formulate an algorithm to determine the optimal value of N. To overcome this difficulty, we apply a heuristic approach: For fixed N, the optimization problem

$$J_{\bar{\omega}}(c) := l(\mathcal{N}_{\bar{\omega},c}; y) + \sum_{n=1}^{N} \phi(|c_n|), \tag{9}$$

is well-equipped to solve by traditional gradient-based optimization method. Now we need to find a way to heuristically determine the optimal $\bar{\omega}$ and the optimal number N. To this end, we introduce the following theorem as a sufficient condition.

Theorem 4.6. *Assuming ϕ satisfies the Assumption 3.3 and Assumption 3.4. Let $\bar{\mu}$ be a finitely supported measure $\bar{\mu} := \sum_{n=1}^{N} \bar{c}_n \delta_{\bar{\omega}_n}$ such that*

1. *$\bar{c}_n \in (\mathbb{R}\backslash 0)^N$ is a local minimum of $J_{\bar{\omega}}$,*
2. *For all $\omega \in \Omega\backslash\{\bar{\omega}_n\}_{n=1,\dots,N}$, it holds $|\bar{p}(\omega)| \lneqq \alpha$, where $\bar{p} = \mathcal{N}^*(\mathcal{N}_{\bar{\omega},\bar{c}} - y) + \sum_{i=1}^{d} (\partial_{x_i}\mathcal{N})^*(\partial_{x_i}\mathcal{N}_{\bar{\omega},\bar{c}} - \partial_{x_i} V)$.*

Then $\bar{\mu}$ is a local minimum of J.

Proof. For arbitrary $\epsilon > 0$ we observe the open ϵ-ball $B_\epsilon(\bar{\omega}_n)$ around a support point of $\bar{\mu} \in M(\Omega)$ and define

$$\Omega_\epsilon := \bigcup_{n=1}^{n=N} B_\epsilon(\bar{\omega}_n)$$

$\mu \in M(\Omega_\epsilon)$ admits the form:

$$\mu = \mu_0 + \tilde{\mu} \quad \text{with} \quad \mu_0 = \sum_{n=1}^{N} c_n \delta_{\bar{\omega}_n}, \quad \tilde{\mu} = \sum_{n} \tilde{c}_n \delta_{\tilde{\omega}_n} + \mu_{\text{cont}}$$

where

$$\{\tilde{\omega}_n\}_{n=1}^{\infty} = \operatorname{atom} \mu \backslash \operatorname{atom} \bar{\mu},$$

$$c_n = \mu(\{\bar{\omega}_n\}), \tilde{c}_n = \tilde{\mu}(\{\tilde{\omega}_n\}).$$

We want to prove for all $\mu \in B_\epsilon(\bar{\mu})$, it holds $J(\bar{\mu}) \leq J(\mu)$. First we observe the quadratic fidelity term, it holds

$$\begin{aligned}\frac{1}{2}\|\mathcal{N}\mu - y\|^2_{L^2(D,\nu)} &\geq \frac{1}{2}\|\mathcal{N}_{\bar{\omega},c} - y\|^2_{L^2(D,\nu)} + \langle \mathcal{N}[\mu_0] - y, \mathcal{N}[\tilde{\mu}]\rangle_{L^2(D,\mu)} + \frac{1}{2}\|\mathcal{N}\tilde{\mu}\|^2_{L^2(D,\nu)} \\ &\geq \frac{1}{2}\|\mathcal{N}_{\bar{\omega},c} - y\|^2_{L^2(D,\nu)} + \langle \mathcal{N}[\mu_0 - \bar{\mu}] - y, \mathcal{N}\tilde{\mu}\rangle_{L^2(D,\mu)} + \langle \mathcal{N}\bar{\mu} - y, \mathcal{N}\tilde{\mu}\rangle_{L^2(D,\nu)} \\ &\geq \frac{1}{2}\|\mathcal{N}_{\bar{\omega},c} - y\|^2_{L^2(D,\nu)} - \|\mathcal{N}\|^2\epsilon\|\tilde{\mu}\|_{M(\Omega)} + \langle \bar{p}, \tilde{\mu}\rangle.\end{aligned}$$

For the penalty term, it holds

$$\Phi(\mu) = \Phi(\mu_0) + \Phi(\tilde{\mu}) = \sum_{n=1}^{N} \phi(|c_n|) + \sum_n \phi(|\tilde{c}_n|) + \int_\Omega d|\mu_{\text{cont}}|.$$

Choosing $c_n = \bar{c}_n$, the cost thus satisfies

$$J(\mu) \geq J_{\bar{\omega}}(\bar{c}) - \epsilon\|\mathcal{N}\|^2\|\tilde{\mu}\|_{M(\Omega)} + \underbrace{\alpha\Phi(\tilde{\mu}) + \langle \bar{p}, \tilde{\mu}\rangle}_{=(1)}.$$

For (1) we estimate

$$\begin{aligned}\Phi(\tilde{\mu}) + \langle \bar{p}, \tilde{\mu}\rangle &= \int_\Omega \alpha + \bar{p}(\omega)d|\mu|_{\text{cont}} + \sum_n \alpha\phi(|\tilde{c}_n|) + \bar{p}(\tilde{w}_n)\tilde{c}_n \\ \Big(|\bar{p}| \leq (1-\delta)\alpha\Big) &\geq \sum_n \alpha\phi(|\tilde{c}_n|) - (1-\delta)|\tilde{c}_n| \\ \Big(\text{Assumption 3.4(1)}\Big) &\geq \sum_n \alpha\delta|\tilde{c}_n| - \frac{\gamma_1|\tilde{c}_n|^2}{2},\end{aligned}$$

since $\tilde{c}_n$ is small, the l^2 term is dominated by the l^1 norm, we have

$$\Phi(\tilde{\mu}) + \langle \bar{p}, \tilde{\mu}\rangle \geq (\alpha\delta - \frac{\alpha\gamma_1}{2})\|\tilde{c}\|_{l^1} \geq (\alpha\delta - \frac{\alpha\gamma_1}{2})\|\tilde{\mu}\|_{M(\Omega)}.$$

Putting the estimates together, we have

$$J(\mu) \geq J(\bar{\mu}) - \left(\epsilon\|\mathcal{N}\|^2 + (\alpha\delta - \frac{\alpha\gamma_1}{2})\right)\|\tilde{\mu}\|_{M(\Omega)}$$

Choose ϵ such that $\epsilon\|\mathcal{N}\|^2 + (\alpha\delta - \frac{\alpha\gamma_1}{2}) = 0$ we get the result. □

5. ALGORITHMS

In this section, we introduce the optimization algorithm. In the first step, we generate the data for training and validation. In the second step, we insert neurons. And finally we trained the network with nonconvex penalty to get a sparse network. The process is repeated a fixed time.

Denote $\mathcal{W}(D)$ as the space of functions that can be represented by measures with bounded variation, i.e.,

$$\mathcal{W}(D) := \left\{ f \in L^2(D,\nu) : \quad f(x) = \mathcal{N}[\mu] \text{ for some } \mu \in M(\Omega) \right\},$$

with,

$$\|f\|_{W(D)} := \min_{\mu \in M(\Omega)} \|\mu\|_{M(\Omega)} \quad \text{such that } f = \mathcal{N}[\mu].$$

The algorithm is divided into 3 parts:

(1) data generation
(2) greedy insertion
(3) redundant neurons elimination

5.1. **data generation.** We come back to the original problem (P), the problem can be written as

$$\inf_{u,y} J(y,u)$$

s.t.

$$E(y,u) = 0,$$

with

$$E(y,u) = \begin{bmatrix} \frac{d}{dt}y - \Big(f(y) + g(y)u\Big) \\ y(0) - x \end{bmatrix}.$$

Assuming the y can be written the function of u, define $U := L^2(t_0, T)$, $U_{ad} := \{u \in U : \hat{E}(u) := E(y(u), u) = 0\}$, we observe the reduced problem

$$\inf_{u \in U_{ad}} \hat{J}(u)$$

We assume the function $y(u)$ of u is well-defined, the first variation at $\bar{u}$ evaluated at direction $u \in U_{ad}$ is

$$\langle \hat{J}_u(\bar{u}), u\rangle_{U^*,U} = \langle y_u^*(\bar{u}) \cdot J_y(y(\bar{u}), \bar{u}) + J_u(y(\bar{u}), \bar{u}), u\rangle_{U^*,U},$$

where we denote $*$ as the adjoint operator. Since $y(u)$ is implicitly defined in general and the derivative is difficult to solve, we leverage the Lagrange argument and implement the adjoint approach. Define the space $\mathcal{E} := \mathrm{ran}(E)$ and $p \in \mathcal{E}^*$. Observe

$$L(p, y, u) = J(y, u) + \langle p, E(y, u)\rangle_{\mathcal{E}^*,\mathcal{E}}.$$

Taking the partial derivative w.r.t. y, it holds

$$\langle L_y(y,u), d\rangle_{Y^*,Y} = \langle J_y, d\rangle_{Y^*,Y} + \langle E_y^*(y,u)p, d\rangle_{Y^*,Y} \quad \text{for all } d \in Y,$$

At the optimal pair (y^*, u^*) we have $L_y(y^*, u^*) = 0$. The corresponding costate p^* satisfies

$$p^* = -E_y^{-*} \cdot J_y \quad \text{in } \mathcal{E} \tag{10}$$

where $E_y^{-*} := (E_y^*)^{-1}$. Plug p^* in the Lagrangian to get the gradient w.r.t. u:

$$\langle L_u(y^*, u^*, p^*), u\rangle_{U^*,U} = \langle J_u(y^*, u^*, p^*), u\rangle_{U^*,U} + \langle E_u^*(y^*, u^*) \cdot p^*, u\rangle_{U^*,U}$$

$$= \langle 2\beta u^* - g(y^*)p^*, u\rangle_{U^*,U}.$$

p^* can be calculated by (10) and integration by parts:

$$\begin{cases} p_t^* + \partial_y(f + g\cdot u)^T p^* = -l_y \\ p^*(T) = 0 \end{cases}$$

For the implemented data-driven approach, it is essential to have the causality-free initial dataset $\{x^j, V(0,x^j), \nabla V(0,x^j)\}_{j=1}^N$, where by causality-free we mean the sample are generated without approximated valuation at nearby points. [8] provides a comprehensive survey about generating the initial dataset. In [1] an adjoint approach is used to compute the optimal value and the reduced gradient (Barzilai-Borwein updates).

5.2. **greedy insertion.** In [2], a conditional gradient algorithm for finitely many samples is presented. To avoid the constraint, like in [12] we implement the stereographic projection:

$$\Psi : \mathbb{S}^d \to \mathbb{R}^d, \quad z \mapsto \left(\frac{2z}{1+\|z\|^2}, \frac{1-\|z\|^2}{1+\|z\|^2}\right).$$

Denote K the number of the samples, the optimal neural net representation of the true target function is supported on at most K atoms.

In each iteration f_t, we calculate $\bar{f}_t$ such that

$$\bar{f}_t \in \arg\min_{f\in\mathcal{W}(D)} \langle f, J'(f_t)\rangle_{L^2(D,\nu)}.$$

Denote $g = -J'(f_t) \in L^2(D,\nu)$ for $f_t(x) = \int_\Omega \varphi_\omega^t(x) d\mu_{f_t}(\omega)$. The conditional gradient algorithm relies on solving at each iteration the "Frank-Wolf step", namely, we maximize:

$$\Omega \ni \omega \mapsto |p_t(\omega)| = \left|\frac{1}{K}\sum_{k=1}^K \sigma(x_k;\omega) g_t(x_k)\right|. \tag{11}$$

To find the global maximal is a NP-hard problem. However heuristic approaches are available for greedy insertion.

5.3. **nonconvex penalized algorithm.** [TBD] as in Pieper.

6. Generalization bound and convergence rate

$$L(\hat{f}) - \inf_{f\in L^2(D)} L(f) \le \left[\inf_{f\in L^2(D)} L(f) - \inf_{f\in\mathcal{W}(D)} L(f)\right] + 2\sup_{f\in\mathcal{W}(D)} |l(f) - L(f)| + |l(\hat{f}) - \inf_{f\in\mathcal{W}(D)} l(f)|. \tag{12}$$

7. numerical Example

7.1. **Van der Pol oscillator.** We consider the optimal control of the Van der Pol oscillator expressed as

$$\min_{u\in L^2(0,T;\mathbb{R})} \int_0^T \left(y_1^2(t) + y_2^2(t) + \beta u^2(t)\right) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1-y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

The open loop problem is:

The HJB equation at $t = 0$ is:

$$\partial_t V(0,x) - \frac{1}{4\beta}\nabla V^T(0,x) g(x) g^T(x) \nabla V(0,x) + \nabla V(0,x)^T f(x) + l(x) = 0$$

8. Might be useful later

8.1. **integral representation of convex functionals on the space of measure.** For $f : \Omega \to (-\infty, +\infty]$ a convex function, we define the functional

$$F : L^1(\Omega, \lambda) \to \mathbb{R}$$
$$v \mapsto \int_\Omega f(\frac{d\mu}{d\lambda}).$$

The functional above admits a weak* l.s.c. envelope, i.e.,

$$\liminf_{\mu_n \overset{*}{\rightharpoonup} \mu} F(\mu) \geq \sup_{u\in C(\Omega)} \langle \mu, u\rangle - \int_\Omega f^*(u) d\lambda.$$

Definition 8.1 (the infimal convolution)**.** Let f and g be 2 closed proper convex function on $\mathbb{R}^n$, the infimal convolution of f and g is defined as

$$(f \triangledown g)(v) = \inf_{x\in\mathbb{R}^n} (f(x) + g(v-x)),$$

with $\mathrm{dom}(f\triangledown g) = \mathrm{dom}(f) + \mathrm{dom}(g)$.

Proposition 8.1. *Let f and g be 2 closed proper convex function on $\mathbb{R}^n$, it holds*

$$f \triangledown f^\infty = f,$$
$$g \triangledown g = g \qquad \textit{for } g \textit{ subadditive and } g(0) = 0.$$

We define the indicator function $\mathbb{1}_H(u)$ of a closed and convex set $H \subset C(\Omega)$ with nonempty interior:

$$\mathbb{1}_H(u) = \begin{cases} 0 & u \in H \\ +\infty & u \notin H. \end{cases}$$

The Fenchel conjugate is calculated as

$$\mathbb{1}_H^*(u^*) = \sup_{u \in H} \langle u^*, u \rangle$$

The following result transform the conjugation into infimal convolution.

Theorem 8.2 (infimal convolution in the functional space)**.** *Let Ω be a compact subset of $\mathbb{R}^{d+1}$. Given $C(\Omega)$ with its dual $M(\Omega)$. Let $H \subset C(\Omega)$ closed and convex and denote J the linear functional*

$$J : L^1(\Omega, m) \to \mathbb{R}, \quad u \mapsto \int_\Omega j(u(\omega)) dm(\omega).$$

If every $\mu \in M(\Omega)$ admits the Lebesgue-Nikodym decomposition $\mu = \frac{d\mu}{d\lambda} \cdot \lambda + \mu^s$. Then it holds for $m = |\lambda| + \mu^s$,

$$[J + \mathbb{1}_H]^*(\mu) = \int_\Omega [J^* \nabla \mathbb{1}_H^*]^{**}(\mu) dm = \int_\Omega j^* \nabla \mathbb{1}_\Gamma^* \left(\frac{d\mu}{d\lambda}(\omega) \right) d\mu + \int_\Omega \mathbb{1}_\Gamma^* \left(\mu^s(\omega) \right) d\mu^s,$$

with $\Gamma := \mathrm{cl}\{\mathrm{ran}(u) : u \in H\} \cap \mathrm{dom}(J)$.

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